

Definition 1.5: Parallel Vectors

Two non-zero vectors $\vec{v}$ and $\vec{w}$ ($\vec{v}, \vec{w} \neq \langle 0, 0 \rangle$) are *parallel* (denoted as $\vec{v} \parallel \vec{w}$) if there exists a non-zero scalar $\lambda \in \mathbb{R}$ such that:

$$\vec{w} = \lambda \vec{v}$$